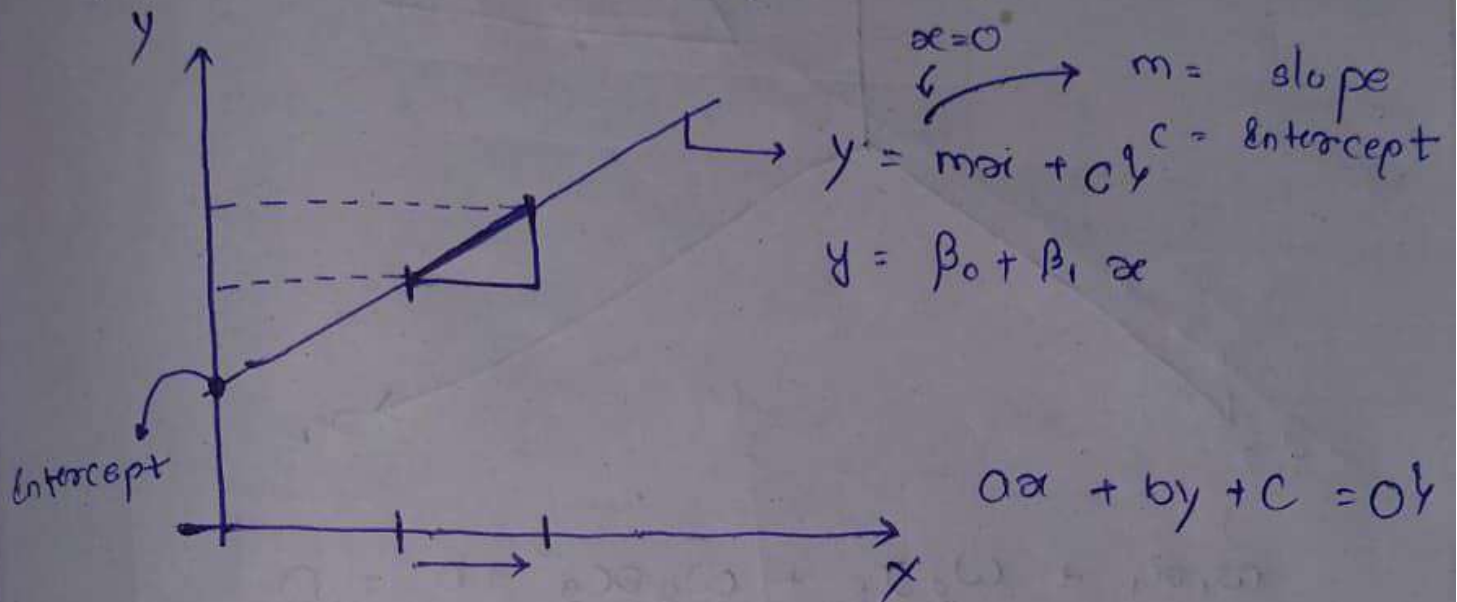
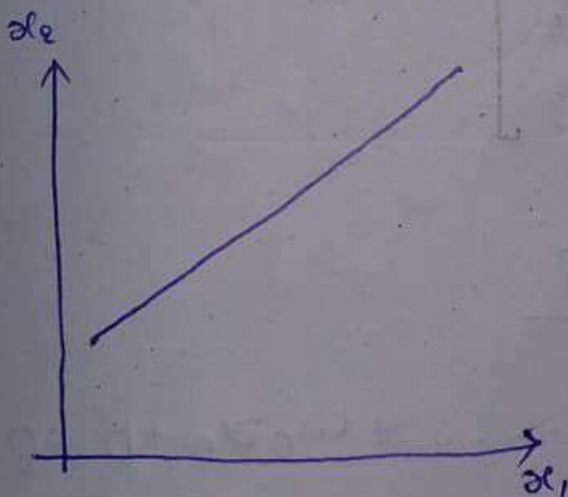


Equation of line, 3d plane and Hyperplane (n Dimension)



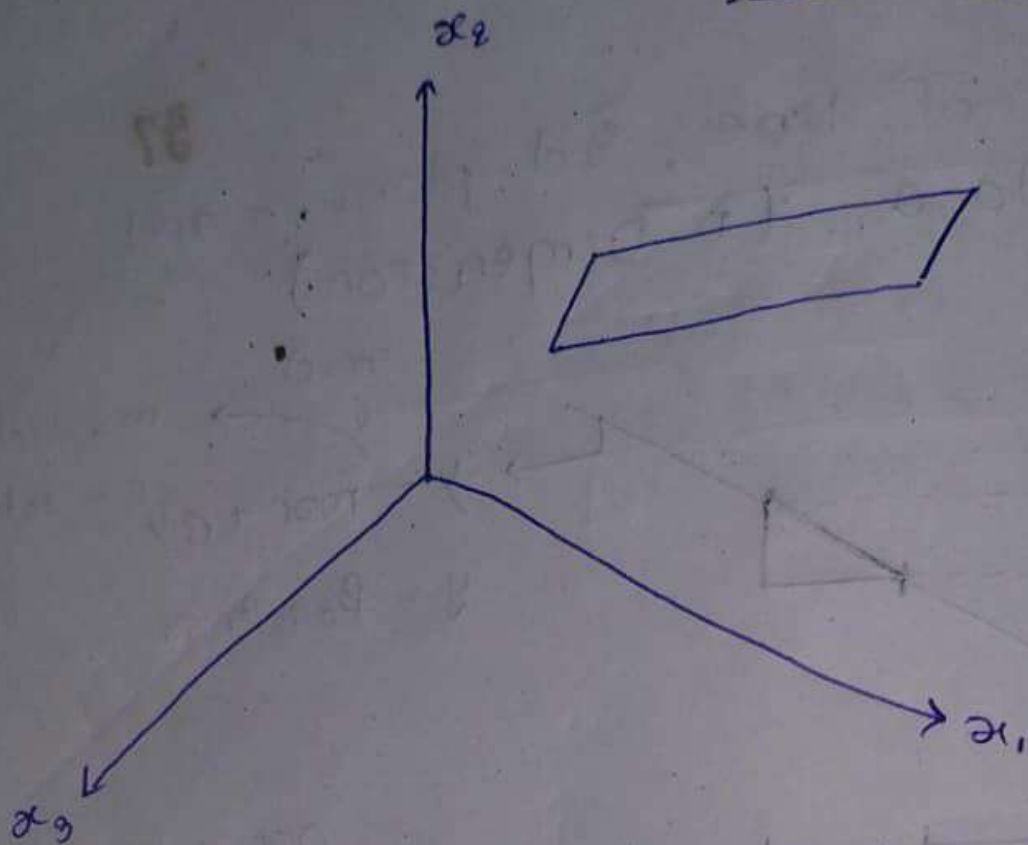
$$\begin{aligned}
 by + c &= -ax \\
 by &= -ax - c \\
 y &= \boxed{\frac{-a}{b}} x - \boxed{\frac{c}{b}} \rightarrow c
 \end{aligned}$$

$\downarrow$
 m



$$\omega_1 x_1 + \omega_2 x_2 + b = 0$$

$$\boxed{\omega^T x + b = 0} \Rightarrow \text{Eq}^n \text{ of a straight line}$$



$$\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + b = 0$$

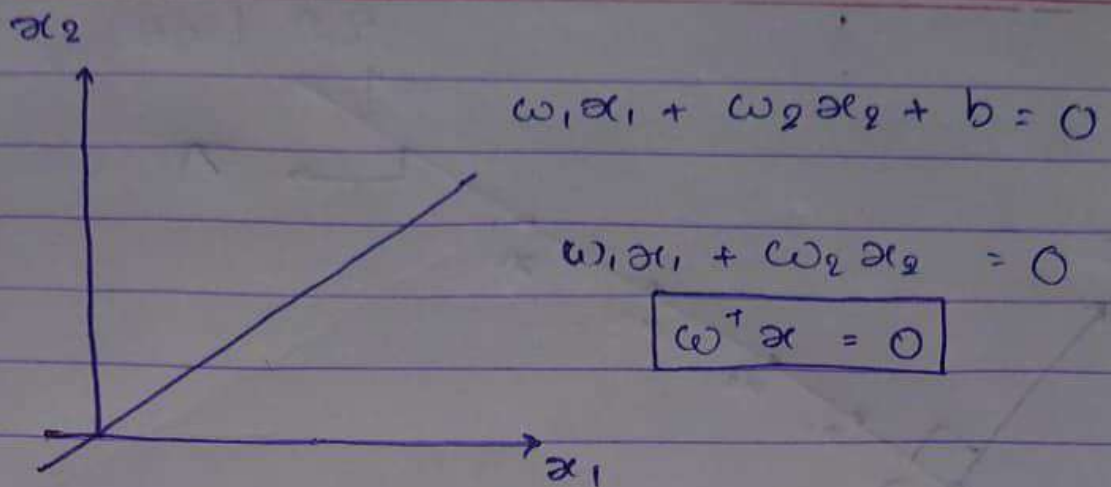
$$\boxed{\omega^T x + b = 0}$$

$$\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \cdot x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

n - Dimensional plane

$$\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + \dots + \omega_n x_n + b = 0$$

$$\boxed{\omega^T x + b = 0}$$

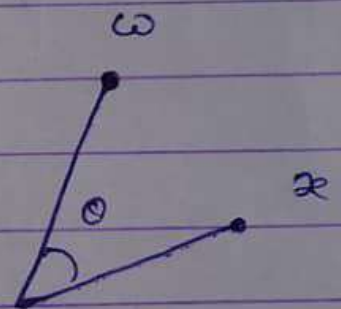


Equation of a straight line passing through an origin

$$w^T x = 0$$

Equation of a plane = $\Pi_n : w^T x = 0$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$



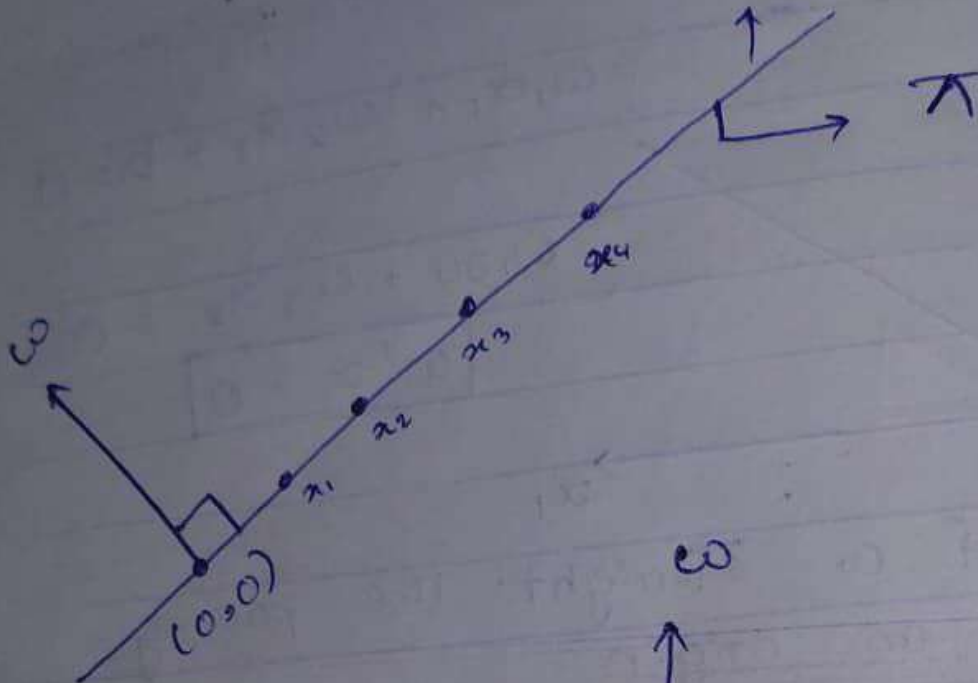
$$w^T x = 0$$

$$w \cdot x = w^T x = \|w\| \|x\| \cos \theta = 0$$

$$\theta = 90$$

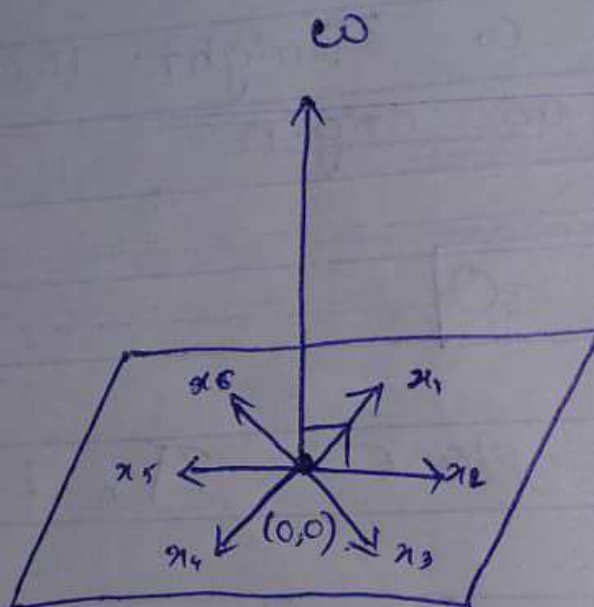
$$\cos \theta = 0$$

2D line



$$\omega \perp \pi$$

intercept = 0



$$\omega^T x = 0$$